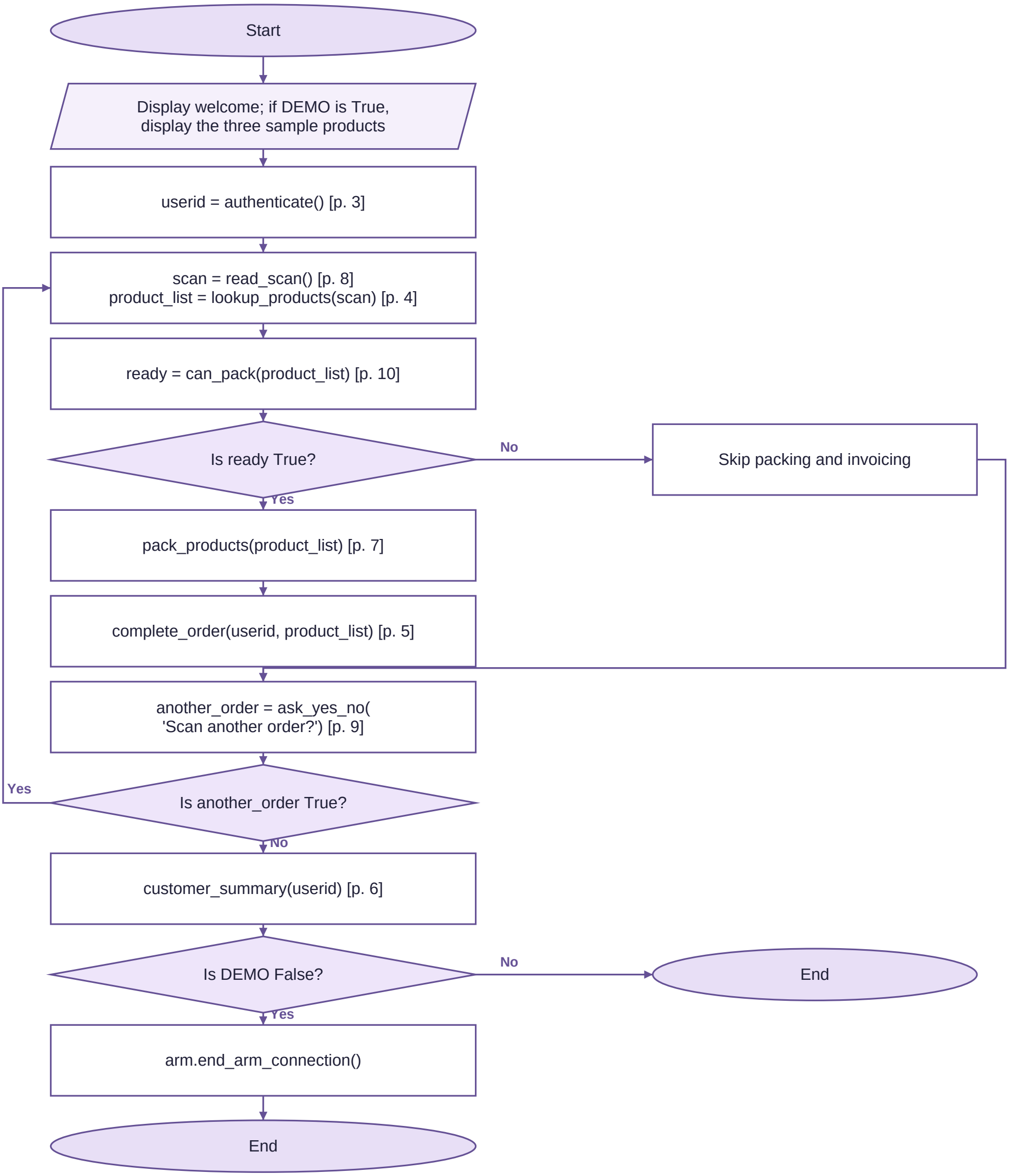
main()

Responsibility Team

Parameters None

Returns None



FUNCTION DETAILS

MANDATORY product_list is a 2D list of [name, numeric price] pairs, shared by packing and invoicing. Packing removes invalid entries from this same list. The invoice uses only the remaining valid entries.

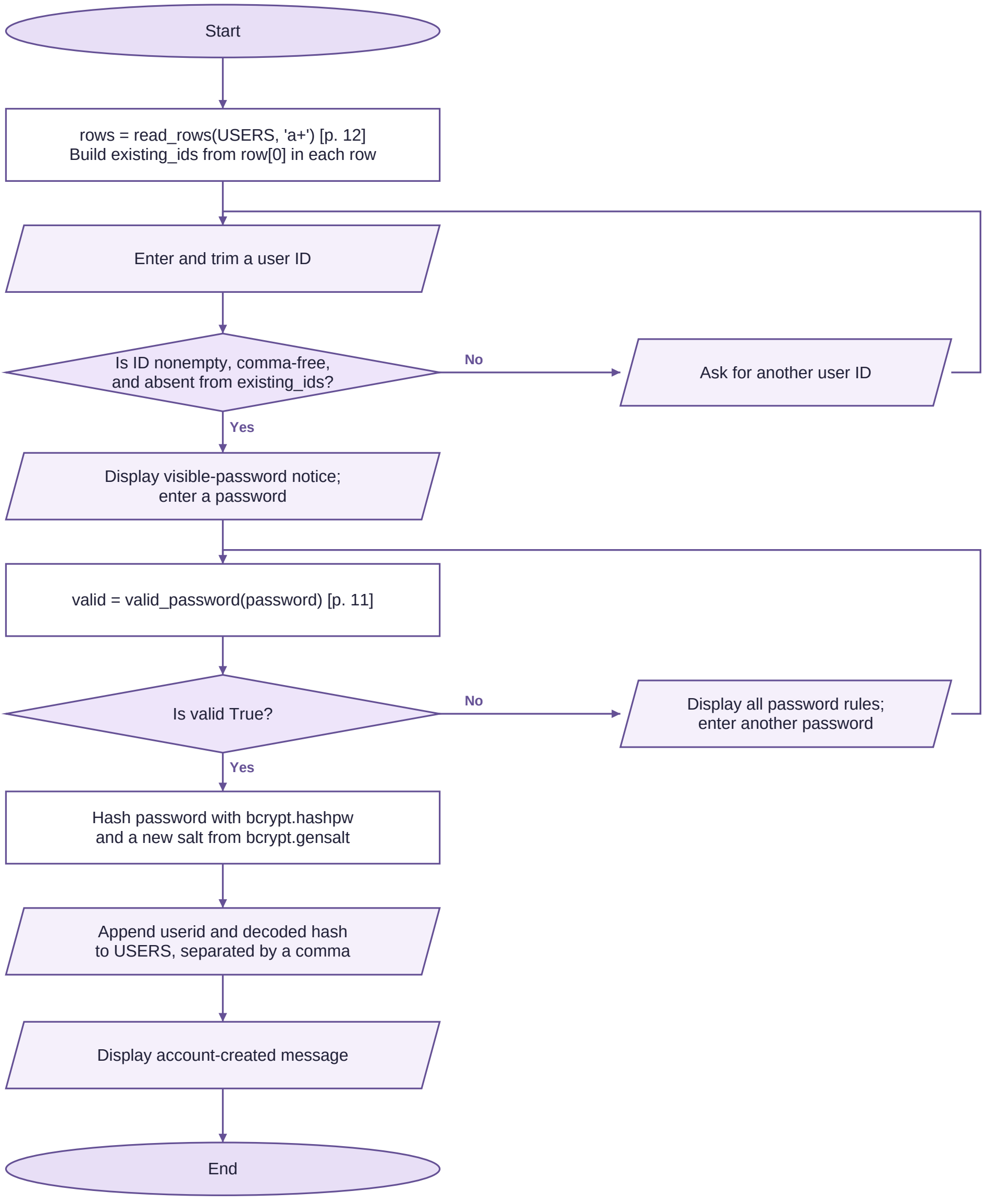
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

sign_up()

Lead Anshi Wang | Support Joleen Sadek | MacID wanga275

Parameters None

Returns None



FUNCTION DETAILS

USERS is users.csv in the lab and demo_users.csv during laptop practice.
The password helper checks length and all four character requirements. Only the hash is stored.

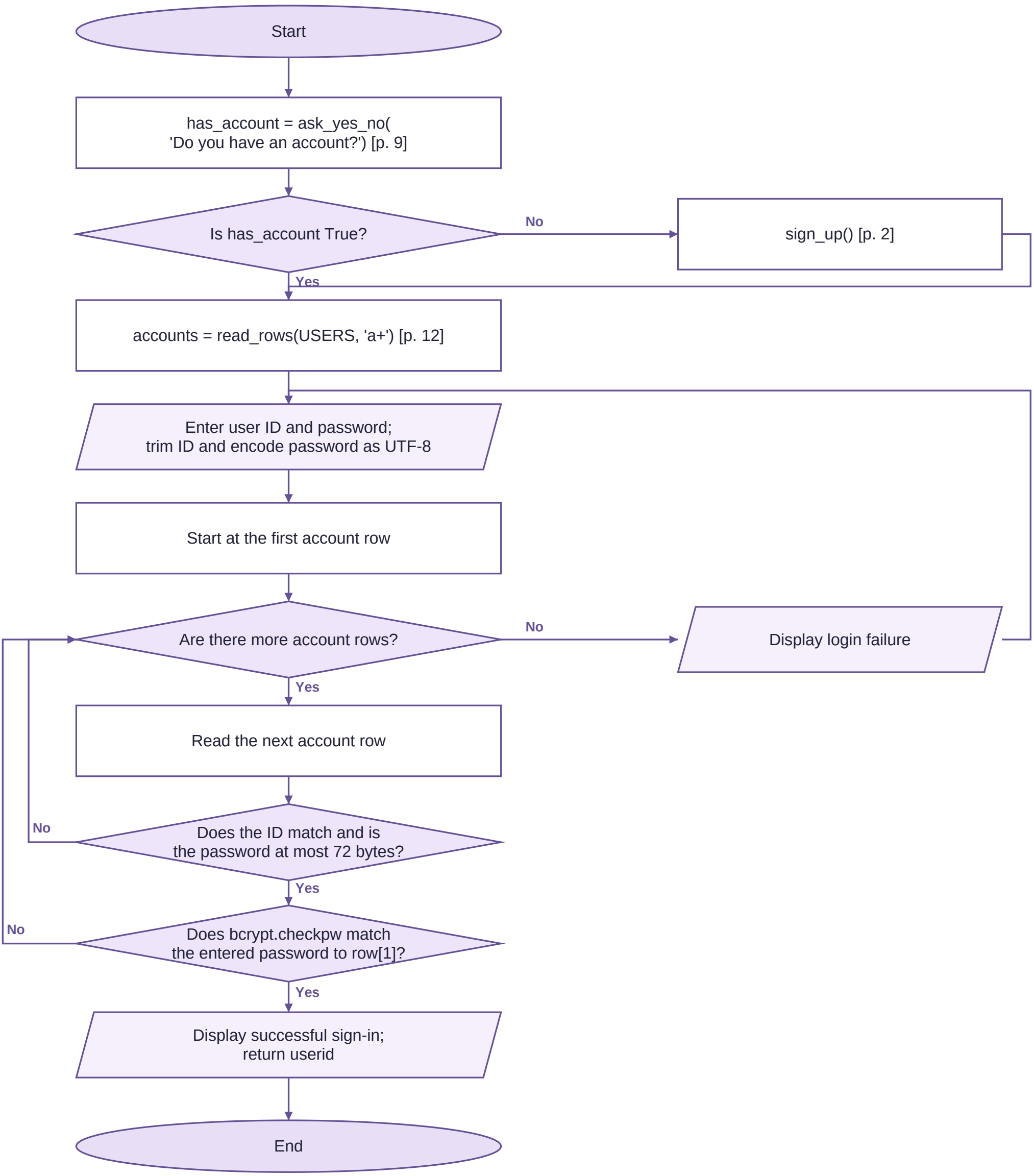
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

authenticate()

Lead Anshi Wang | Support Margarette Pangan | MacID wanga275

Parameters None

Returns userid (string)



FUNCTION DETAILS

A failed attempt repeats the credential prompt. A successful attempt returns the matching ID.
Every account row is checked before a failure message is shown. Passwords remain hashed in USERS.

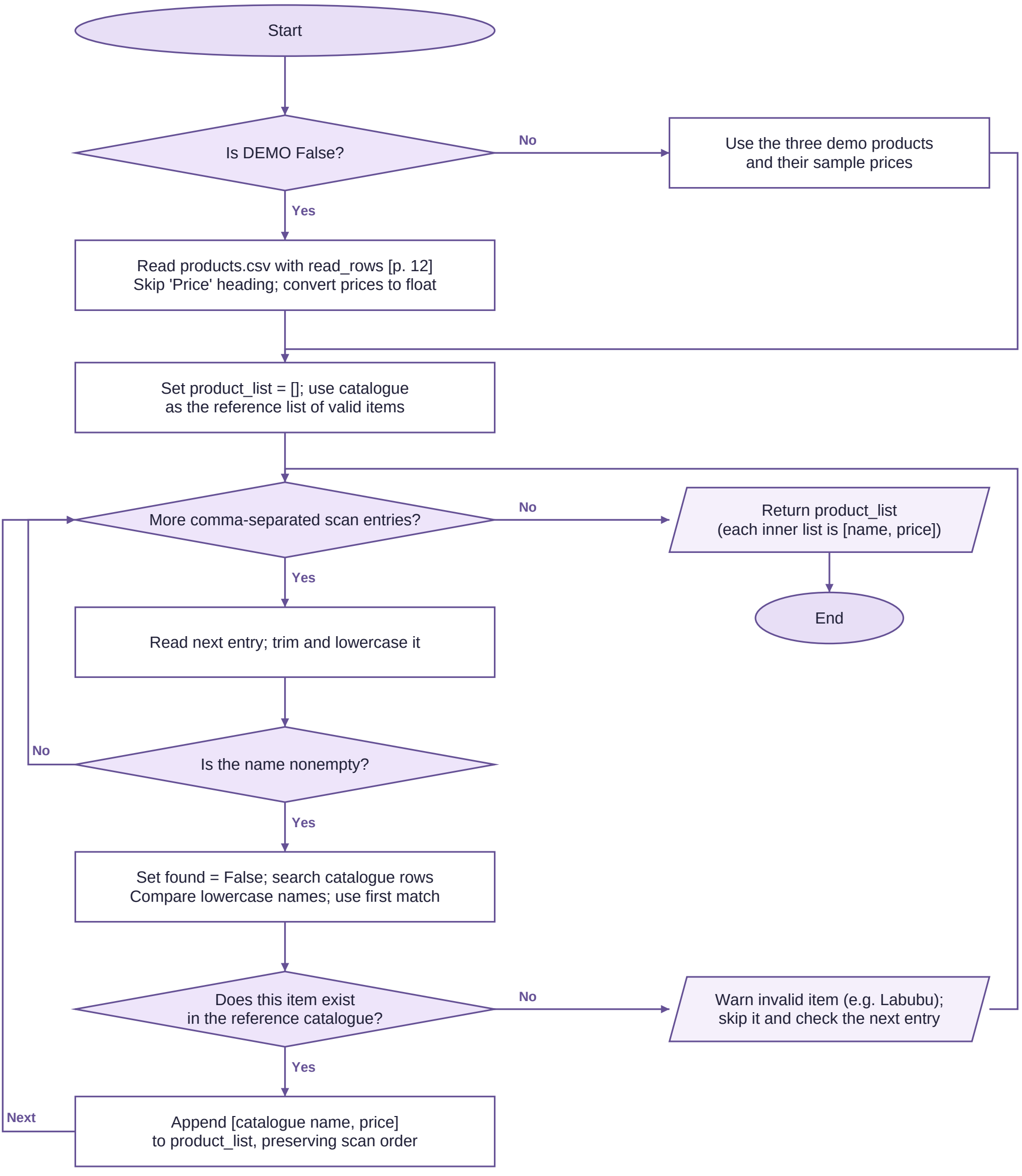
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

lookup_products(products)

Lead Margarette Pangan | Support Anshi Wang | MacID pangam1

Parameters products (comma-separated string)

Returns product_list = [[name, numeric price], ...]



FUNCTION DETAILS

MANDATORY Each inner list contains exactly the product name and its numeric catalogue price.
Use catalogue membership, not a hardcoded Labubu rule. Keep valid items and repeats; all invalid returns [].

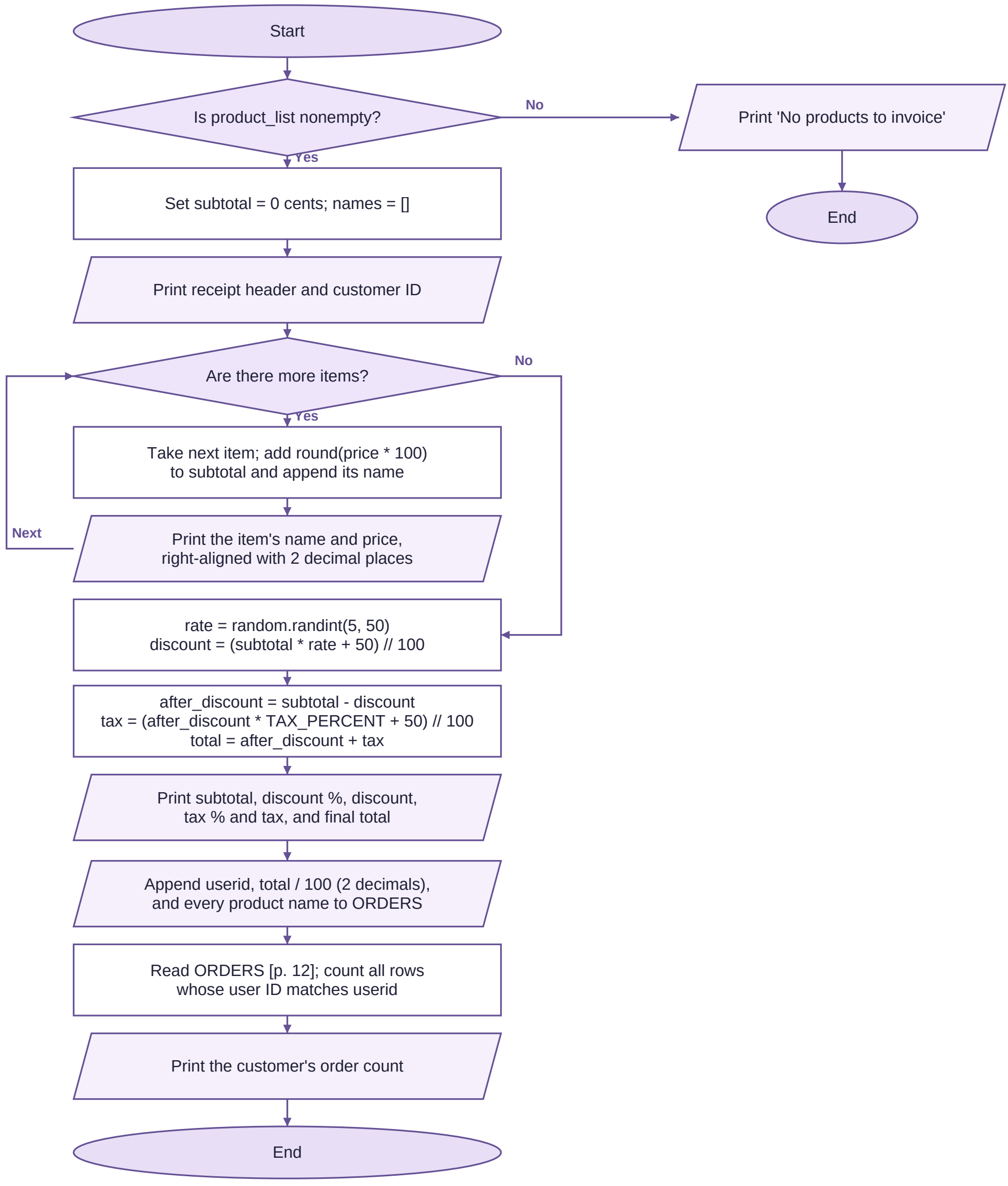
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

complete_order(userid, product_list)

Lead Jose Mella Iturra | Support Anshi Wang | MacID mellaiti

Parameters userid; product_list = [[name, numeric price], ...] (mandatory 2D list)

Returns None



FUNCTION DETAILS

Use the same product_list after packing validation. Invalid items are excluded; an empty list saves no order.
Amounts are calculated in cents; +50 before //100 rounds halves up. TAX_PERCENT = 13 is an assumption.

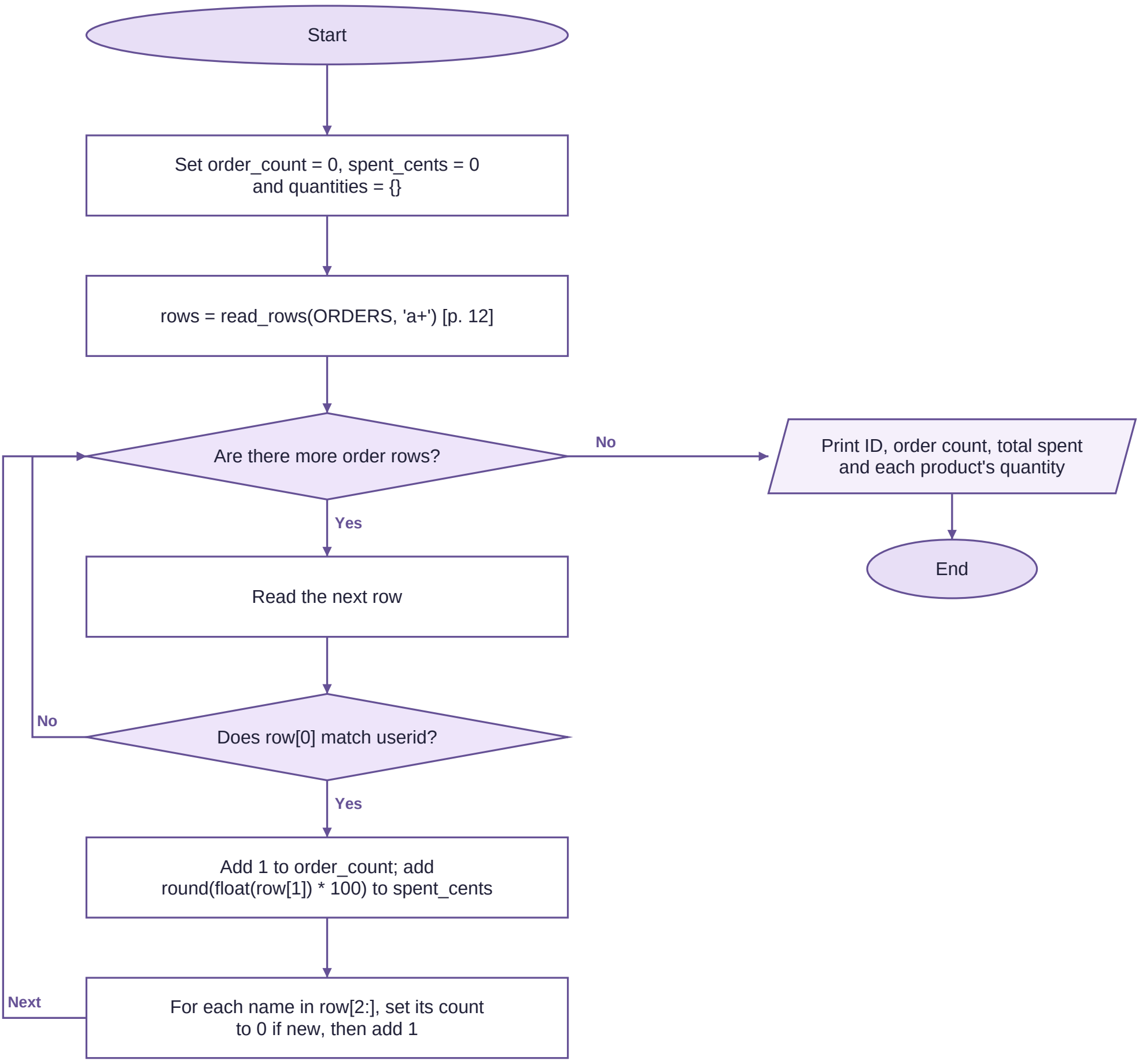
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

customer_summary(userid)

Lead Joleen Sadek | Support Jose Mella Iturra | MacID sadekj3

Parameters userid

Returns None



FUNCTION DETAILS

Totals include only this user. Each occurrence counts once, including duplicates within an order.
Print money to 2 decimals and quantities right-aligned. With zero orders, also print 'No orders yet'.

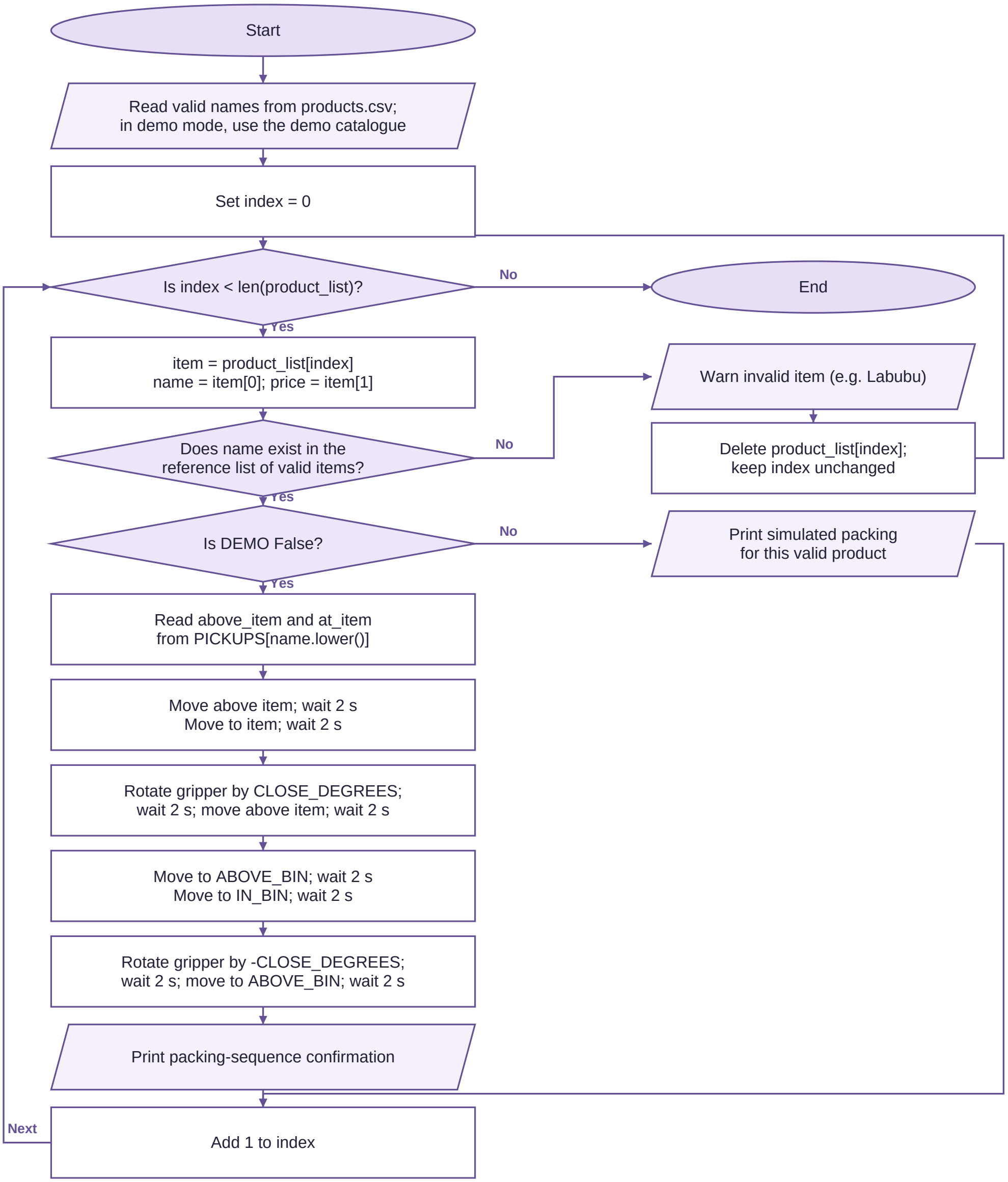
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

pack_products(product_list)

Responsibility Team

Parameters product_list = [[name, numeric price], ...] (mandatory 2D list)

Returns None; invalid entries removed from the shared product_list



FUNCTION DETAILS

Each entry is [name, price]. Check the name before any movement. Valid duplicates are separate units.
The invalid branch removes the entry and keeps the index. The next shifted item is checked; invoicing shares the list.

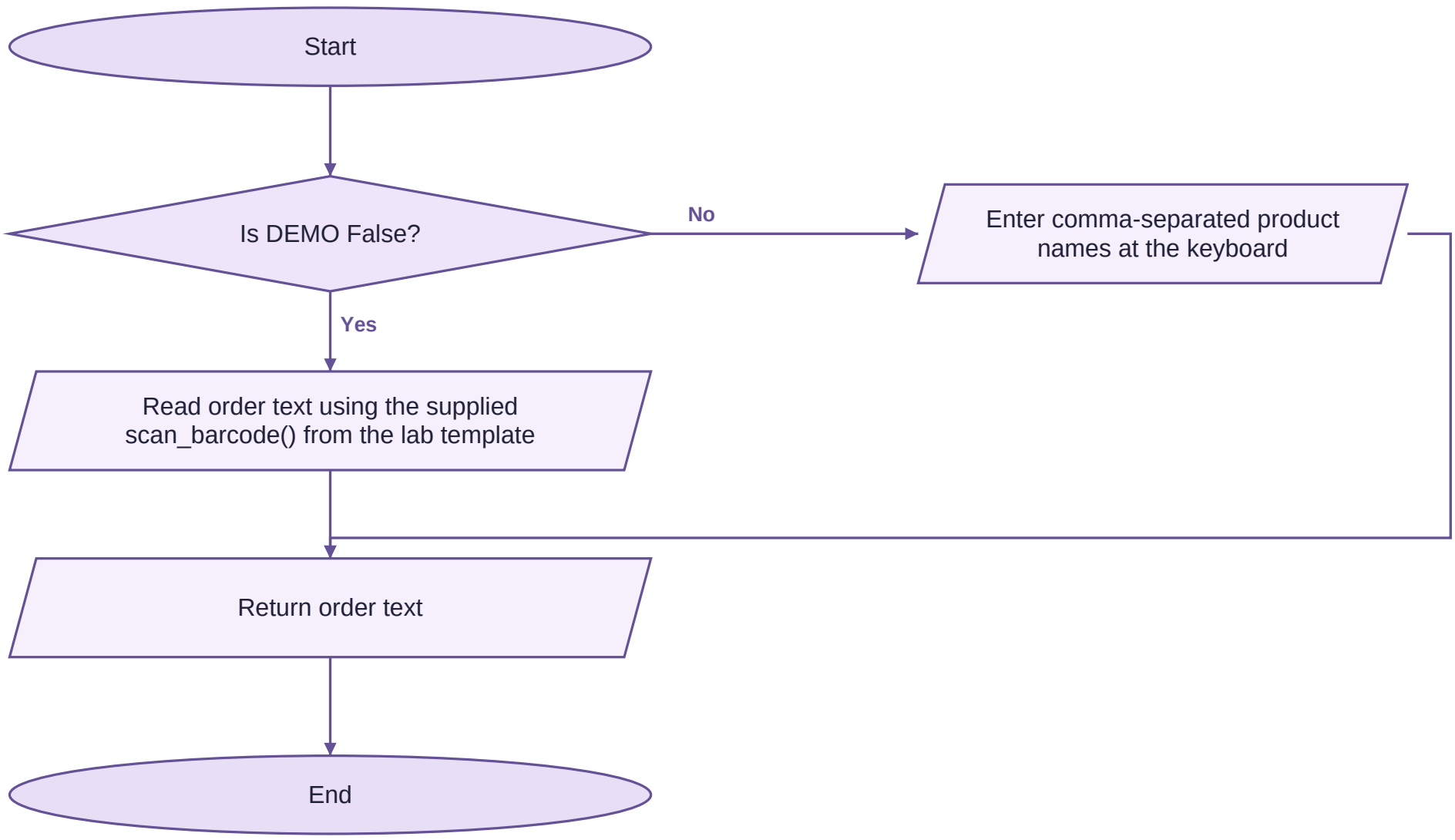
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

read_scan()

Responsibility Team helper; supports the required functions

Parameters None

Returns scan text (string)



FUNCTION DETAILS

This helper selects how the order is entered. The course supplies scan_barcode().
Laptop example: widget,gear,widget. The resulting string is passed to lookup_products().

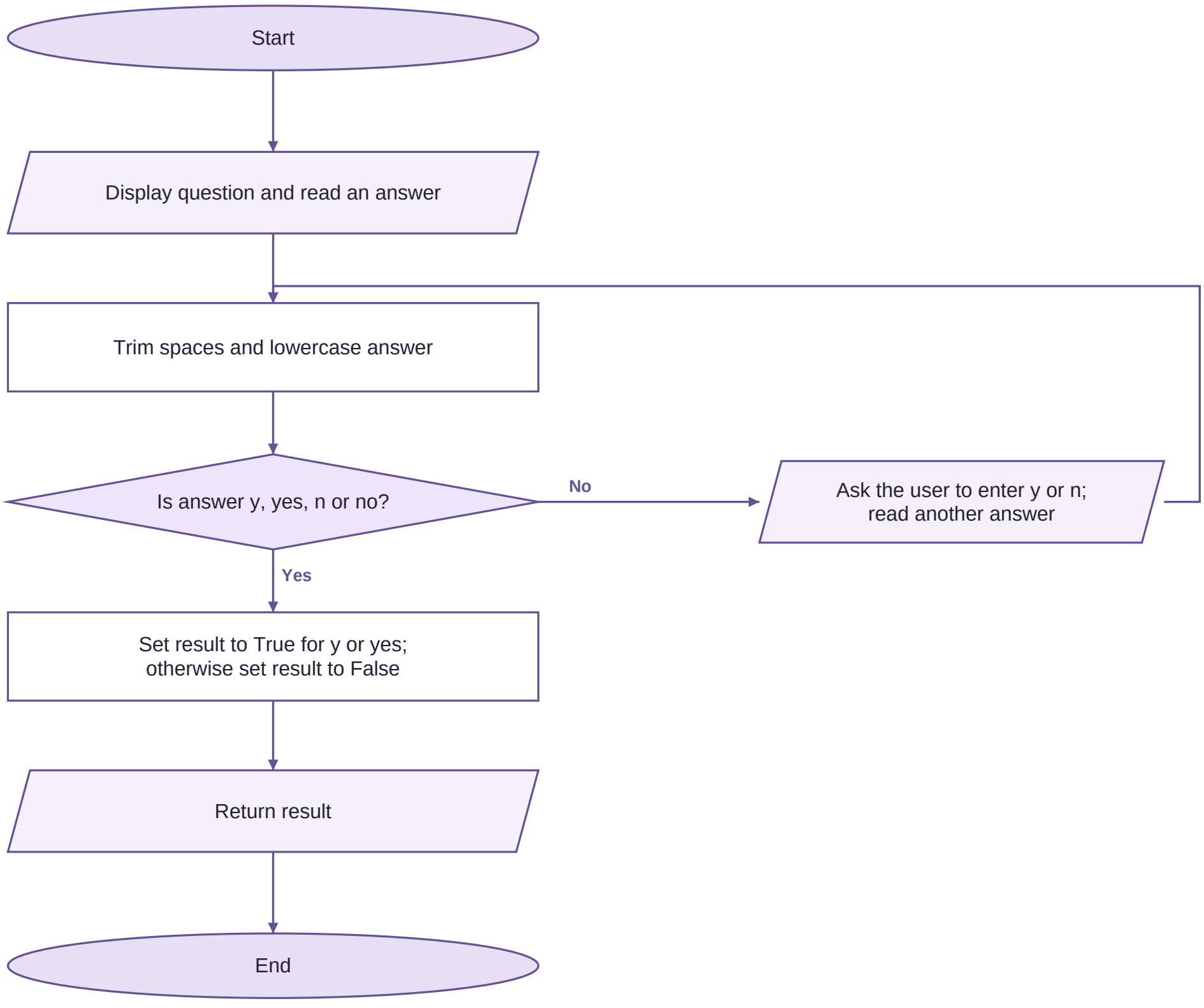
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

ask_yes_no(question)

Responsibility Team helper; supports the required functions

Parameters question (string)

Returns True or False



FUNCTION DETAILS

Used when asking whether the customer has an account and whether to scan another order.
The answer is validated before the caller receives a Boolean result.

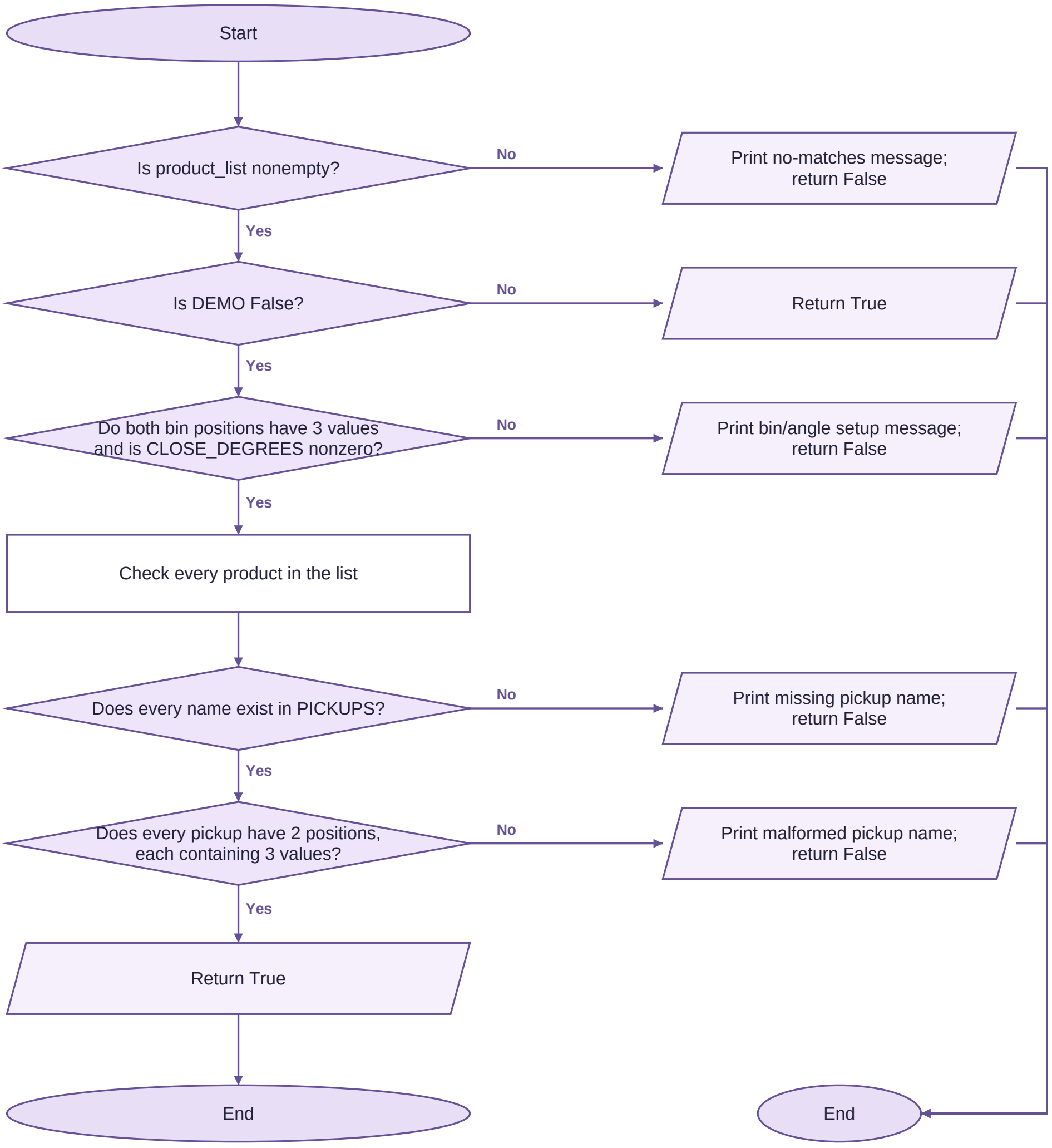
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

can_pack(product_list)

Responsibility Team helper; supports the required functions

Parameters product_list = [[name, numeric price], ...] (mandatory 2D list)

Returns True or False



FUNCTION DETAILS

Per product, check its name first, then its two positions. Stop at the first problem.
PICKUPS checks coordinate availability. Catalogue membership is checked in lookup and again during packing.

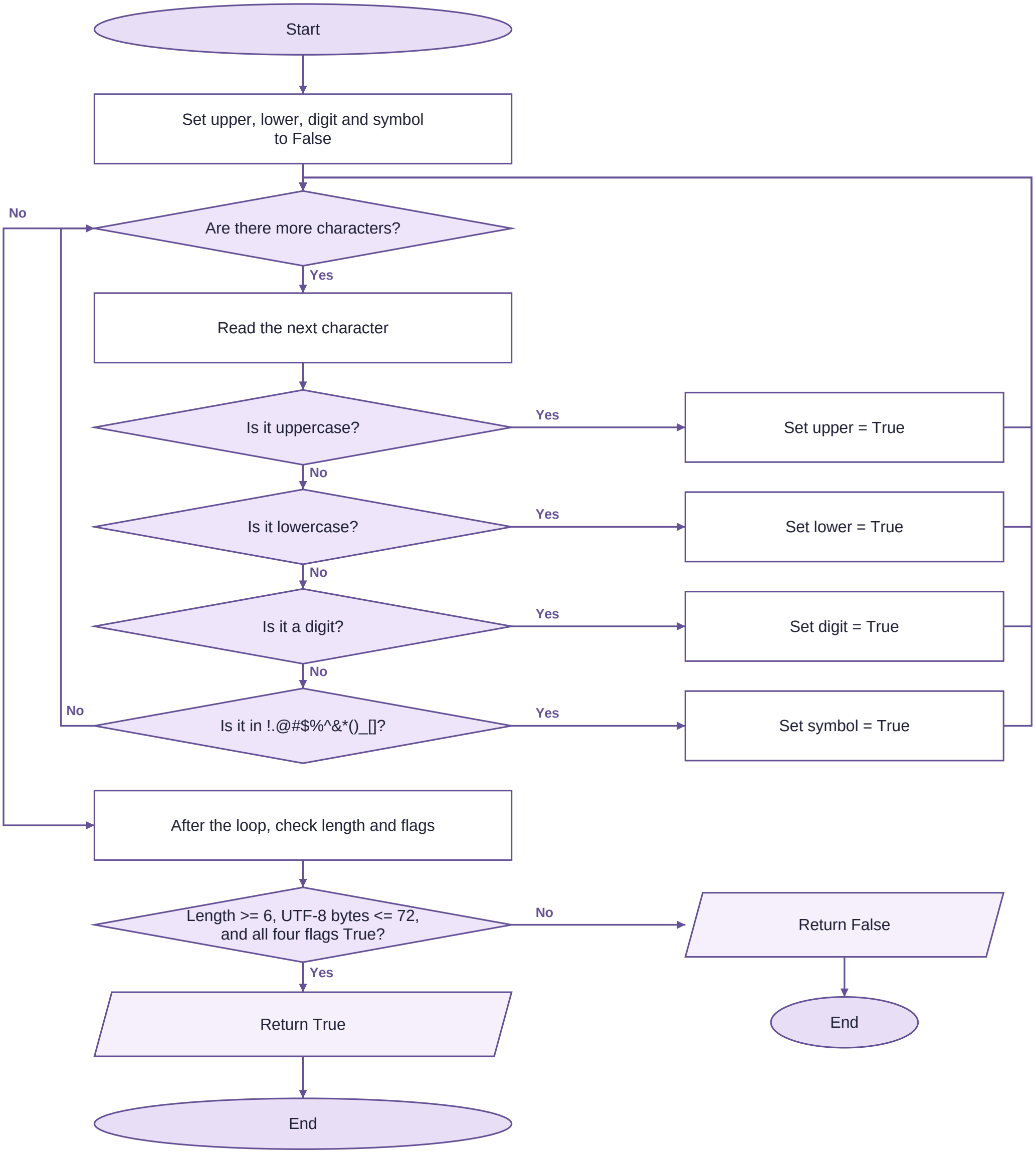
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

valid_password(password)

Responsibility Team helper; supports the required functions

Parameters password (string)

Returns True or False



FUNCTION DETAILS

This helper supports Anshi Wang's sign_up function. All four flags must be True.
The first matching character test sets its flag. A nonmatching character changes no flags.

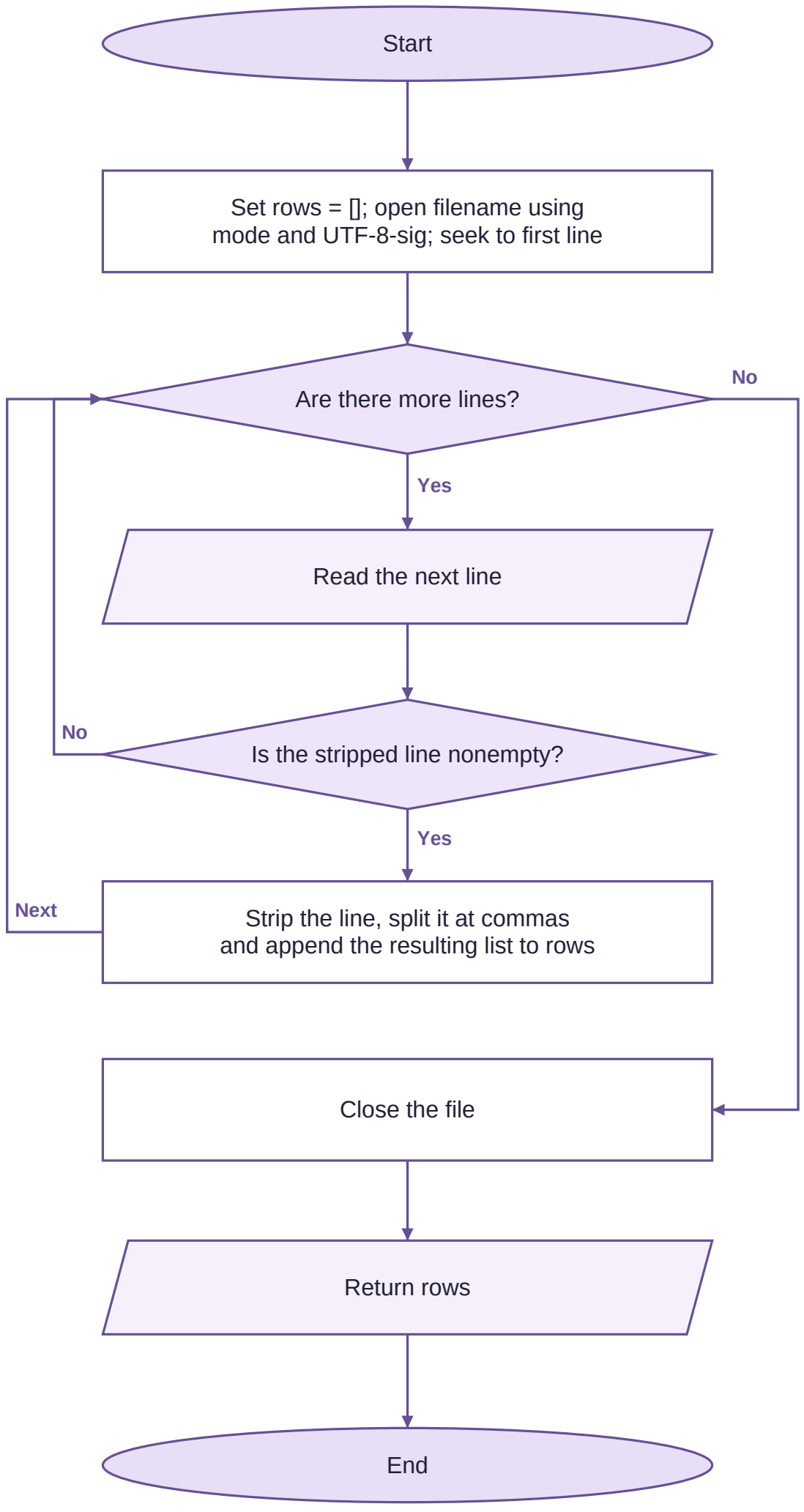
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

read_rows(filename, mode)

Responsibility Team helper; supports the required functions

Parameters filename; mode (default 'r')

Returns list of rows, each a list of strings



FUNCTION DETAILS

For users and orders, mode 'a+' creates a missing file. products.csv is opened for reading. Blank lines are ignored. The course data format uses comma-separated fields.

Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision